

# **Application of Information & Communication Technologies**

## **Lecture-14**

# Recap of Lecture 13

- ◆ Application Software
  - Programs made for the user to perform tasks
- ◆ General Characteristics
- ◆ Major Types
- ◆ Word Processing
- ◆ Spreadsheet
- ◆ Database Software

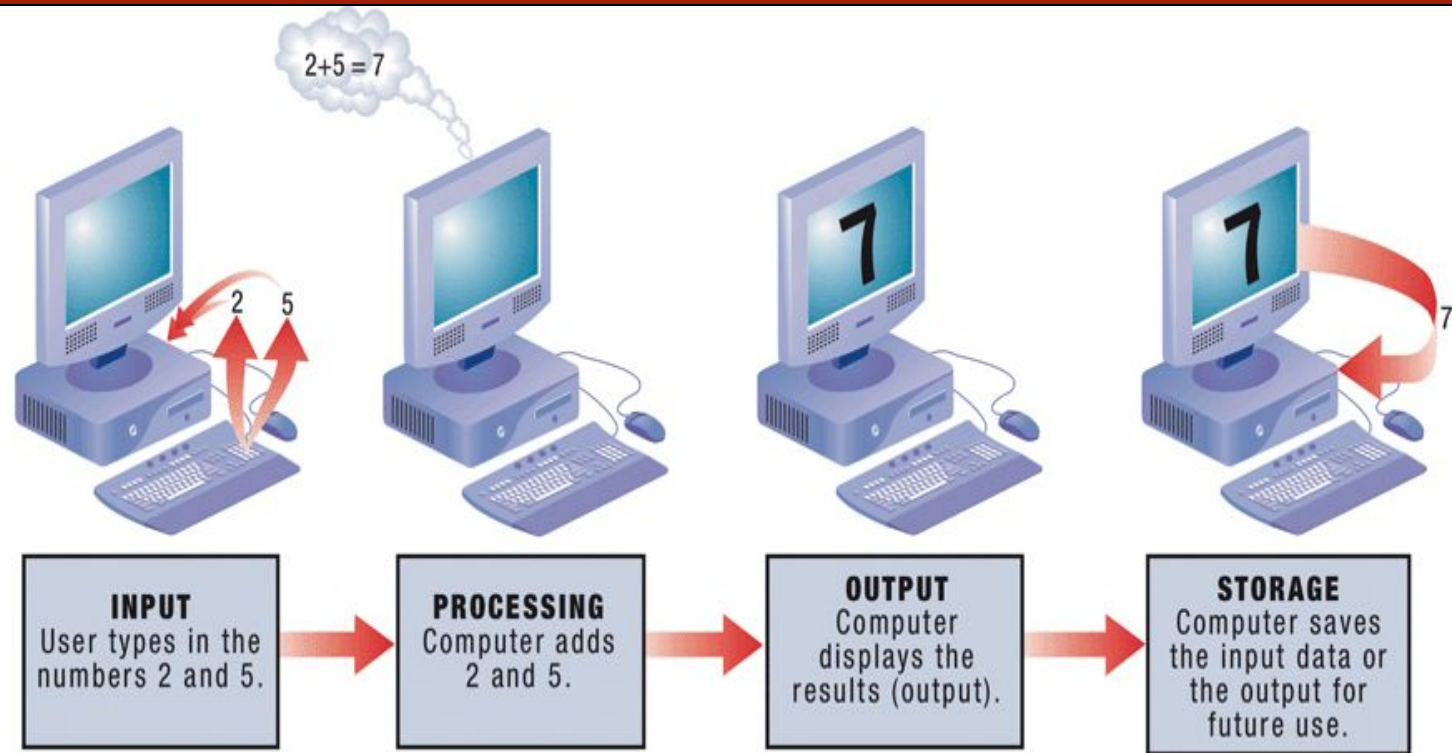
# Overview of Lecture 14

- ◆ Revise for Midterm

# What is a Computer?

- ◆ A computer is an electronic machine.
- ◆ It can take data (input), work on it (processing), show results (output), and save it (storage).
- ◆ A computer only does what a program (set of instructions) tells it to do.

# How a Computer Works: Input, Process, Output, Storage



**FIGURE 1-6**  
The information  
processing cycle.

# Function of Information Technology (IT)

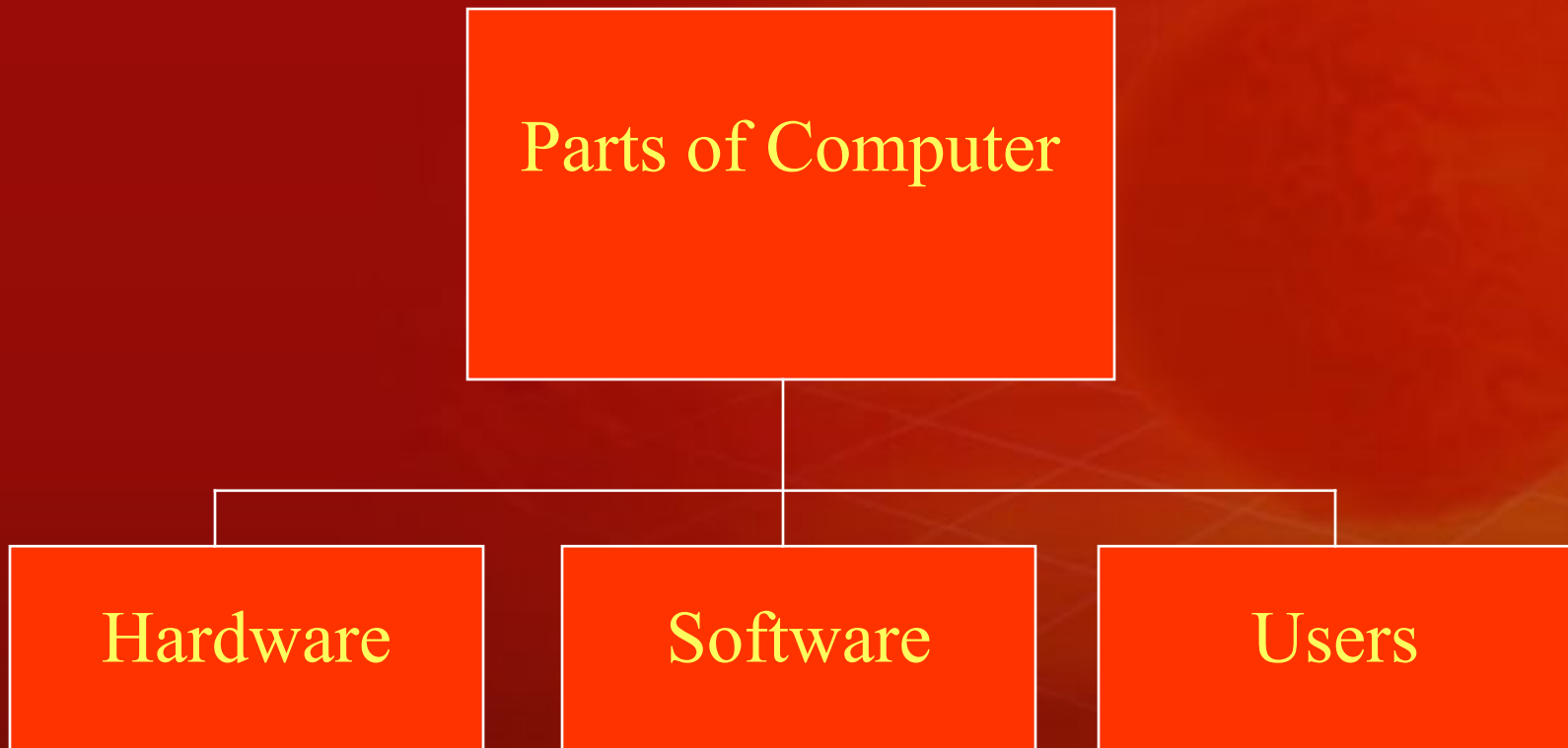
## ◆ Function of IT:

- Input: Collecting data (keyboard, scanner).
- Processing: CPU works on data.
- Storage: Saving data (hard disk, cloud).
- Output: Showing results (monitor, printer).
- Communication: Sharing info (email, WhatsApp, video calls).

# Data vs Information

- ◆ Data = Raw facts (e.g., numbers, marks, words).
- ◆ Information = Processed, meaningful result (e.g., average marks, report).

# Parts of Computer System





# Parts of Computer System

- ◆ **Hardware**: The physical parts of a computer you can touch
  - Internal Hardware
    - Located inside the main box (system unit) of the computer
  - External hardware
    - Located outside the system unit

# Parts of Computer System

- ◆ **Software**: The set of instructions/programs that tell hardware what to do.
  - System Software
    - e.g., Windows, Linux, macOS – helps run the computer.
  - Application Software
    - e.g., MS Word, web browsers, Photoshop, – helps users do tasks.

# Parts of Computer System

- ◆ **Computer Users**: People who use a computer to obtain information
- ◆ Types:
  - End Users (students, teachers, employees).
  - IT Professionals (programmers, system admins).
- ◆ Without users, hardware + software are useless.

# Information Technology (IT)

## ◆ Characteristics of IT:

- Speed
  - Computers process data very fast.
- Accuracy
  - High precision in calculations.
- Automation
  - Tasks run automatically.
- Connectivity
  - Easy to share info via networks/Internet.
- Storage
  - Huge amount of data can be stored in small devices.

# Numbering Systems

- ◆ **Decimal system → Base 10 (digits 0–9)**
- ◆ **Binary system → Base 2 (digits 0 and 1)**
- ◆ **Decimal is human language, Binary is computer language**

# Decimal Number Example

- ◆ Decimal number: 7,216
- ◆ Place values:  $10^3$ ,  $10^2$ ,  $10^1$ ,  $10^0$ 
  - $7 \times 1000 = 7000$
  - $2 \times 100 = 200$
  - $1 \times 10 = 10$
  - $6 \times 1 = 6$
  - 👉 Total = 7,216

## DECIMAL NUMBERING SYSTEM

Each place value in a decimal number represents 10 raised to the appropriate power.

The decimal number  
7,216



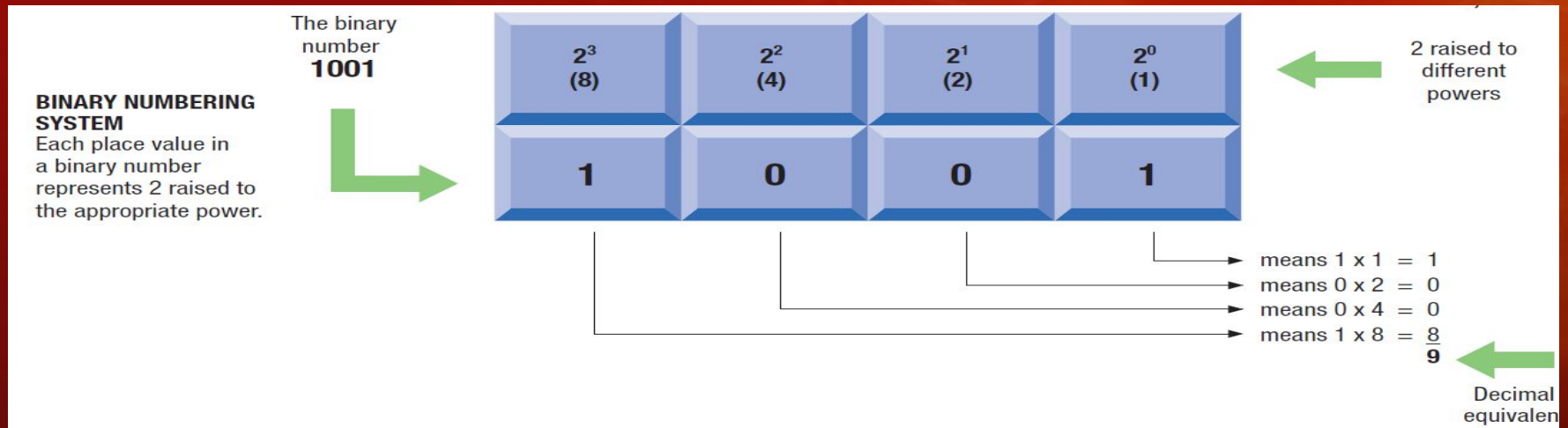
| $10^3$<br>(1,000) | $10^2$<br>(100) | $10^1$<br>(10) | $10^0$<br>(1) |
|-------------------|-----------------|----------------|---------------|
| 7                 | 2               | 1              | 6             |

10 raised to different powers

|                 |   |              |
|-----------------|---|--------------|
| means 6 x 1     | = | 6            |
| means 1 x 10    | = | 10           |
| means 2 x 100   | = | 200          |
| means 7 x 1,000 | = | 7,000        |
|                 |   | <u>7,216</u> |

# Binary Number Example

- ◆ Binary number: 1001
  - Place values:  $2^3$ ,  $2^2$ ,  $2^1$ ,  $2^0$
  - $1 \times 8 = 8$
  - $0 \times 4 = 0$
  - $0 \times 2 = 0$
  - $1 \times 1 = 1$
  - 👉 Total = 9 (decimal)



# Key Point

- ◆ **Decimal: every position = power of 10**
- ◆ **Binary: every position = power of 2**
- ◆ **Same method, different base**

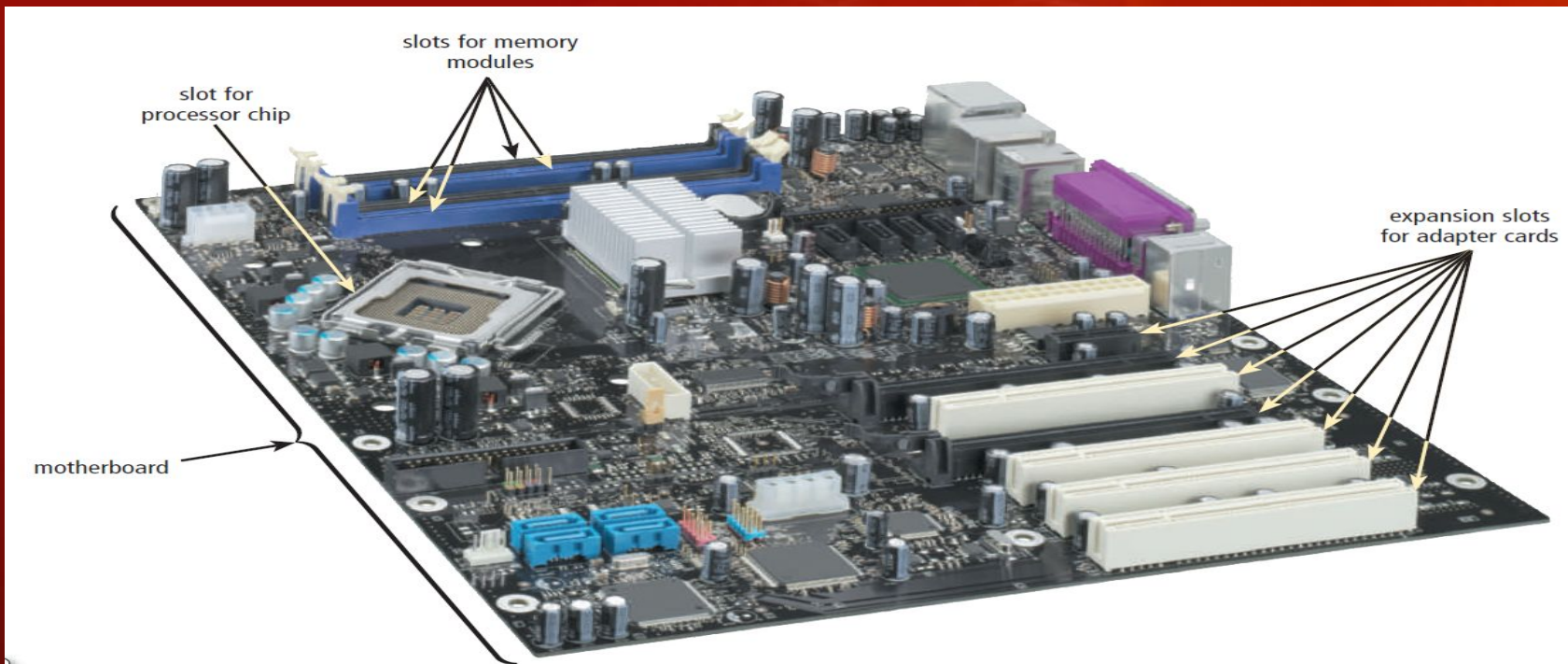


# What is the System Unit?

- ◆ **Main case/box of the computer.**
- ◆ **Holds CPU, memory, storage, power supply, ports.**
- ◆ **Works as the central body of the computer.**

# The Motherboard

- ◆ Largest circuit board.
- ◆ Connects all components.
- ◆ Holds: CPU socket, RAM slots, expansion slots, chipset.



# Power Supply

- ◆ **Power Supply Unit (PSU):** Converts (AC) wall electricity → usable computer power (DC).
- ◆ Includes cooling fan (sometimes variable speed).
- ◆ Some peripherals use external AC adapters (e.g., printer, modem).



# Memory

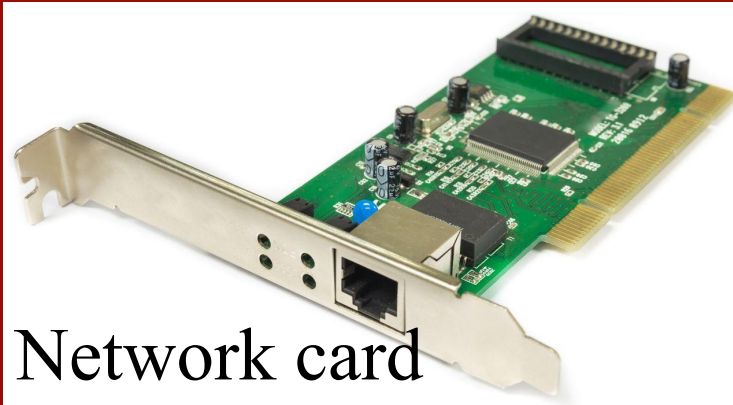
- ◆ **RAM (Temporary, fast).**
- ◆ **ROM (Permanent, startup instructions).**
- ◆ **Cache (Very fast, close to CPU).**
- ◆ **Flash (USB, SSD).**

# Drive Bays vs Expansion Slots & Cards

| Feature  | Drive Bays                                       | Expansion Slots                         |
|----------|--------------------------------------------------|-----------------------------------------|
| Location | In the system unit case                          | On the motherboard                      |
| Purpose  | Hold <b>storage devices</b>                      | Add new <b>functionality</b>            |
| Type     | Rectangular <b>openings</b>                      | Connector <b>slots</b>                  |
| Example  | Hard drive, SSD, Optical disc drive, Card reader | Graphics card, Sound card, Network card |



# Type of Expansion Cards



Network card

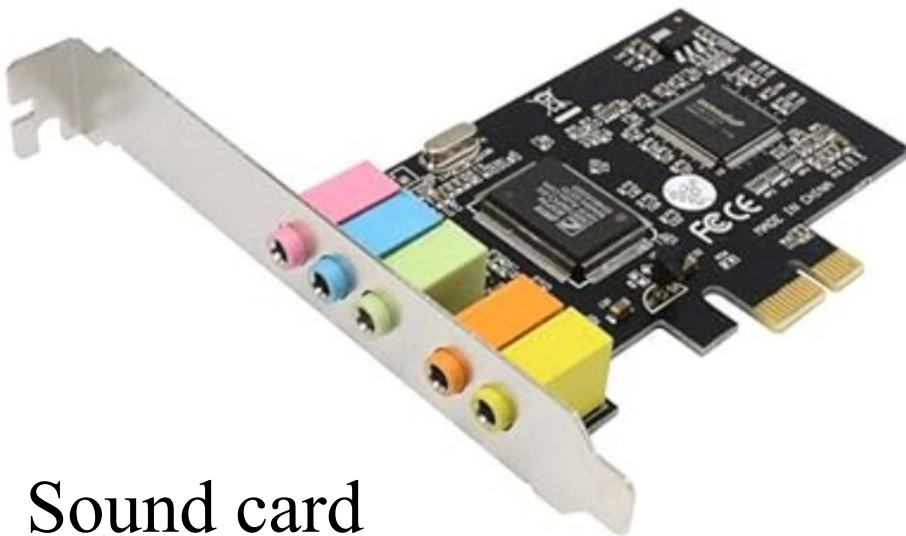
Wireless  
Network cards



Graphics card



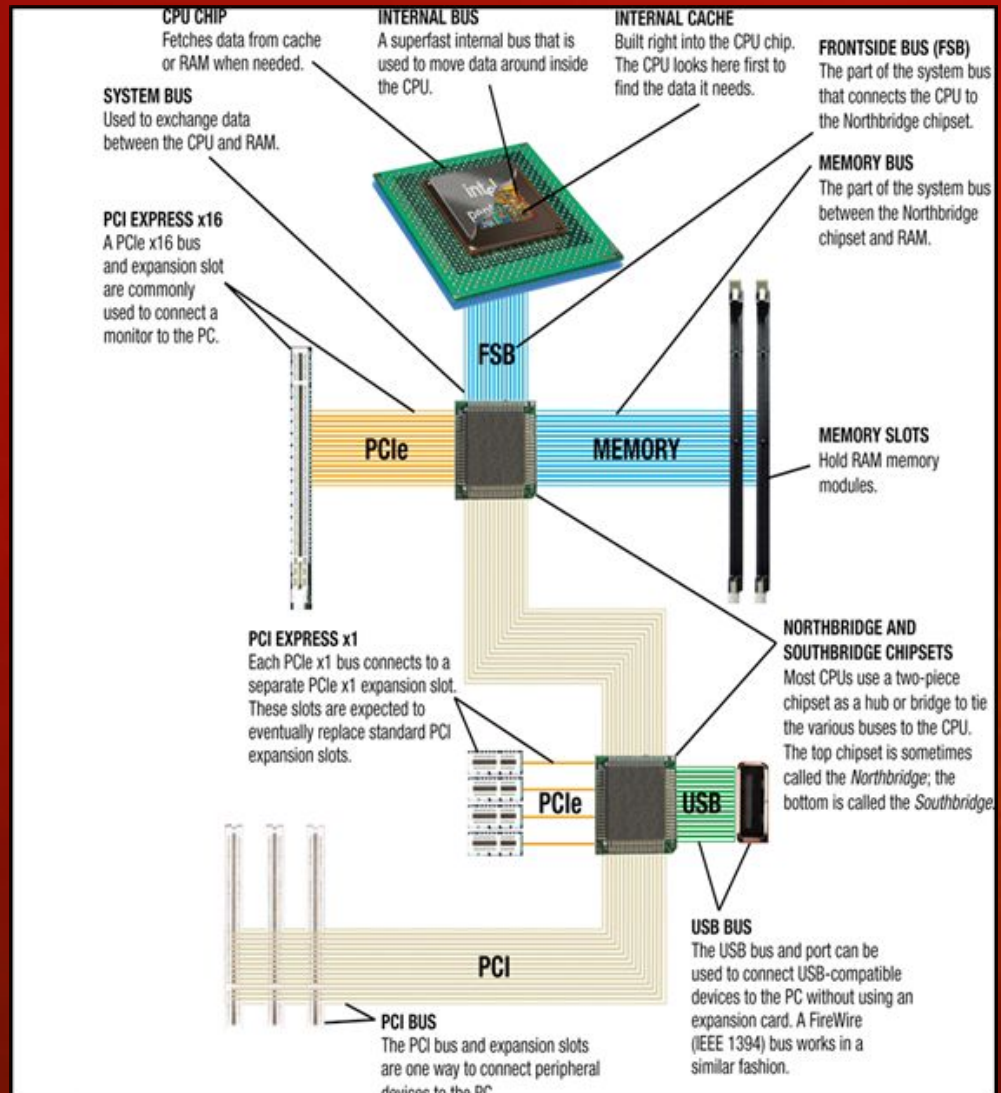
Sound card



# Buses

- ◆ **Electronic Highway for data inside computer.**
- ◆ **Used to transfer data, addresses, control signals.**

**FIGURE 2-16**  
Buses and expansion slots. Buses transport bits and bytes from one component to another, including the CPU, cache, RAM, and peripheral devices.



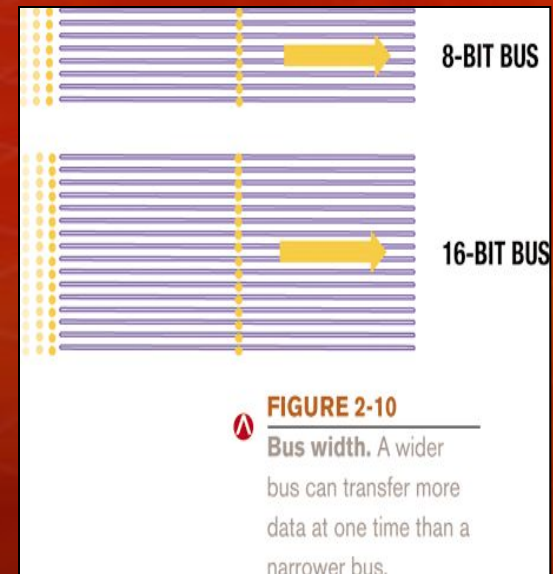
# Type of Buses

- ◆ **Data Bus** → moves actual data
- ◆ **Address Bus** → tells where data goes
- ◆ **Control Bus** → manages signals



# Buses Width

- ◆ **Width** = number of wires/lines.
- ◆ **Wider bus** → more data per cycle
  - (E.g., 8-bit, 16-bit, 32-bit, 64-bit)



# Buses speed

- ◆ **Speed** = how fast data travels.
- ◆ Measured in MHz or GHz.

# Bandwidth

- ◆ **Bandwidth** = Bus Width × Bus Speed.
- ◆ Bus width and speed determine the throughput
- ◆ Example:
  - 32-bit bus × 100 MHz = 400 MB/s.

# System vs Expansion Bus

- ◆ **System Bus** → between CPU & Memory.
- ◆ **Expansion Bus** → connects to peripherals (USB, PCI, etc.).

# Memory and Registers

- ◆ **Stores data and instructions.**
  - **Types: ROM, RAM, Cache, Flash.**
- ◆ **Registers: Very small and high speed memory inside CPU.**
  - **Stores data being used right now.**

# RAM

- ◆ **Random Access Memory.**
- ◆ **Temporary storage for active programs.**
- ◆ **Volatile → erased when power off.**
- ◆ **Why RAM is Important?**
  - **More RAM = faster performance.**
  - **Needed to run modern apps.**

# What is Cache Memory?

- ◆ **Small, very fast memory close to CPU.**
- ◆ **Stores frequently used instructions/data.**
- ◆ **Levels of Cache**
  - **L1 Cache → inside CPU (fastest, smallest).**
  - **L2 Cache → near CPU (bigger, slower than L1).**
  - **L3 Cache → shared by cores.**

# Storage vs Memory

- ◆ Memory (RAM)
  - Temporary, fast, small
  - Loses data when power is off
  - Volatile
  - Example: Working desk (you keep things while working, but clear after)
- ◆ Storage (**HDD/SSD/USB**)
  - Permanent, slower, large
  - Keeps data even when power is off
  - Non-volatile
  - Example: Filing cabinet (data stays safe until removed)



# Access Methods

- ◆ Random Access → data can be read from any place (HDD, SSD). Netflix / YouTube
- ◆ Sequential Access → data read in order (tape storage). Old cassette tapes / VHS
- ◆ Any Real Example ?
- ◆ Which access method is faster?

# Primary Storage (Quick Recap)

- ◆ What is Primary Storage?
  - Directly used by CPU.
  - Includes RAM, Cache, ROM.

# Features of Primary Storage

- ◆ Very fast.
- ◆ Small in size.
- ◆ Expensive.
- ◆ Volatile (RAM, Cache).

# Secondary Storage

- ◆ What is Secondary Storage?
  - Used for long-term storage.
  - Examples: HDD, SSD, Optical discs, Flash drives.

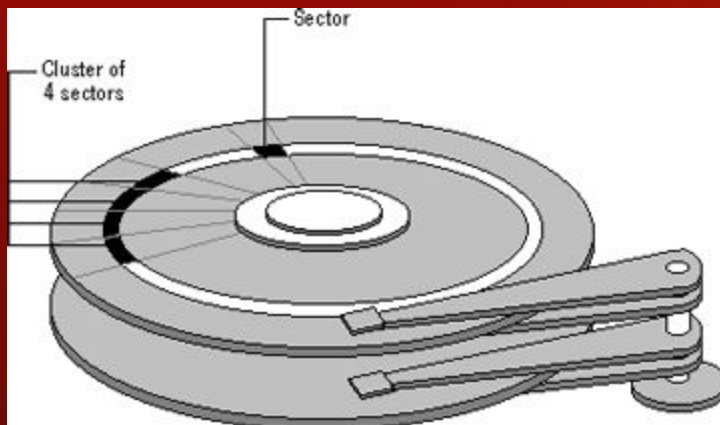
# Hard Disk Drive (HDD)

- ◆ Magnetic storage.
- ◆ Large capacity, cheaper, slower.



# Hard Disk Drive (HDD)

- ◆ Data stored in tracks, sectors, clusters.
- ◆ One or more permanently sealed metal magnetic disks with an access mechanism and read/write heads



# Solid State Drives (SSDs)

- ◆ Use flash memory technology
- ◆ Use less power and have no moving parts
- ◆ Particularly appropriate for portable computers and mobile devices



**FIGURE 3-7**

Solid-state drives (SSDs).



# Optical Discs & Drives

- ◆ **Optical Discs:**
  - A type of storage read from and written to using laser beams
  - Examples: CD, DVD, Blu-ray (BD)
- ◆ Divided into sectors like magnetic discs but use a single spiral track (groove)
- ◆ Used for backup purposes and for storing and transporting music, photos, video, etc.



# What are Input Devices?

- ◆ Devices we use to send data to the computer.
- ◆ Without input → computer cannot work.
- ◆ Examples: keyboard, mouse, scanner, camera, mic.

# How humans give data and instructions to the computer



# What are Output Devices?

- ◆ Devices that show results of computer processing.
- ◆ Convert binary data → human-readable form.
- ◆ Examples: Monitor, Printer, Speakers.

# How computers give results back to us



# System Software

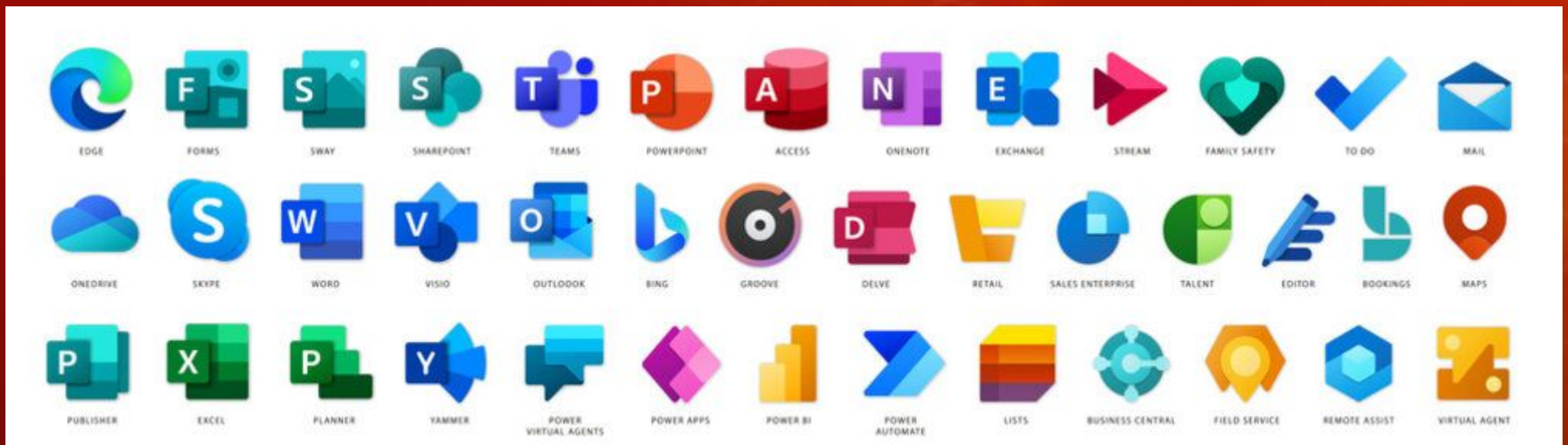
- ◆ Helps the computer itself run.
- ◆ Controls computer hardware.
- ◆ Works in the background.
- ◆ Needed before apps can run.
- ◆ Examples: Windows, Linux, MacOS.





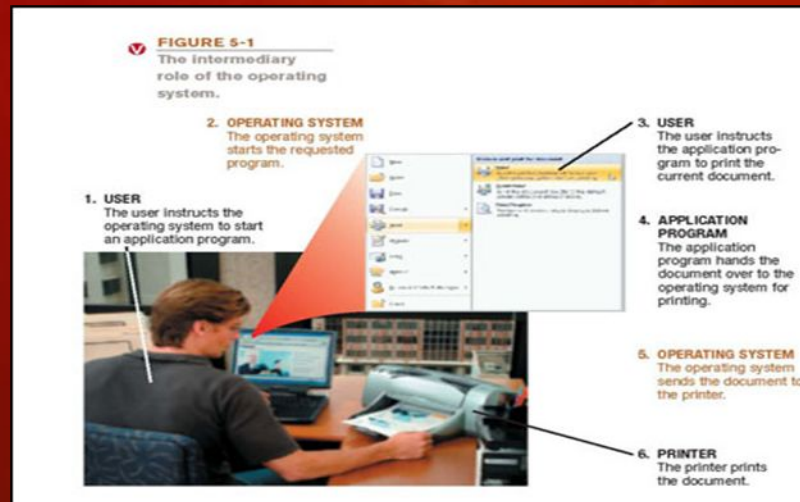
# Application Software

- ◆ Helps the user do tasks.
- ◆ Runs on top of system software.
- ◆ Examples: MS Word, Edge Browser, Map.



# Operating System (OS)

- ◆ A collection of programs that manage and coordinate the activities taking place within a computer system
- ◆ Acts as an intermediary between the user and the computer
- ◆ Main type of system software.
- ◆ Connects: User ↔ Applications ↔ Hardware.
- ◆ Example: Windows, MacOS, Linux, Android.



# Functions of an Operating System

- ◆ **Device Management** (printers, drives).
- ◆ **Memory Management** (RAM usage).
- ◆ **Process Management** (CPU tasks).
- ◆ **File Management** → create, delete, organize.
- ◆ **User Interface** → GUI (desktop) or CLI (commands).
- ◆ **Security** → passwords, permissions.
- ◆ **Multitasking** → run many apps at once.
- ◆ **Communication** → networking, internet access.

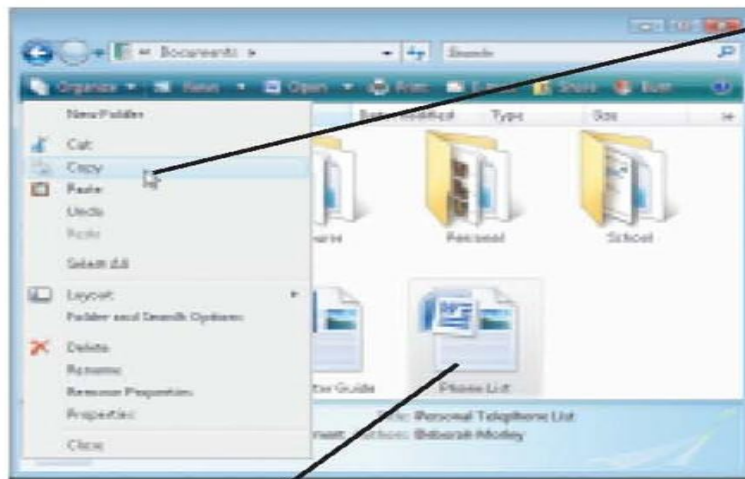


# Utility software

- ◆ Utility software supports the operating system and performs a specific task.
- ◆ Helps in:
  - System performance improvement
  - File management
  - Security and backup
- ◆ Many utilities are built into operating systems (for finding files, viewing images, backing up files, etc.)

# Using Windows Explorer

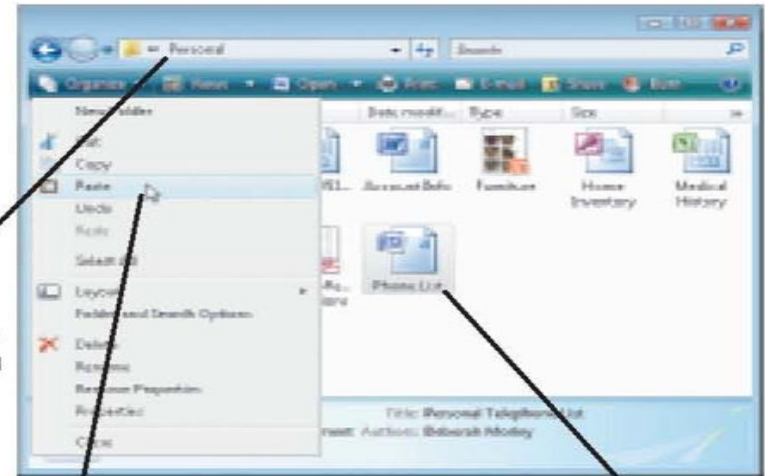
## ◆ Move the file



1. Navigate to the drive and folder containing the file you want to copy or move, and then select the file.

2. Click the Organize button and select *Copy* to copy the file to the Clipboard.

3. Navigate to the drive and folder where you want the file to go.



4. Click the Organize button and select *Paste* to copy the file to the current location.

5. The file is copied.

 **FIGURE 5-20**  
Using Windows  
Explorer to copy files.

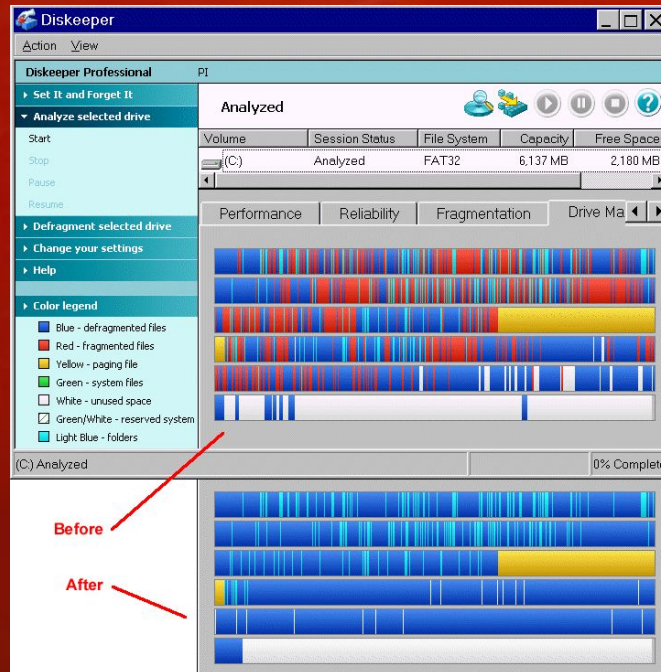
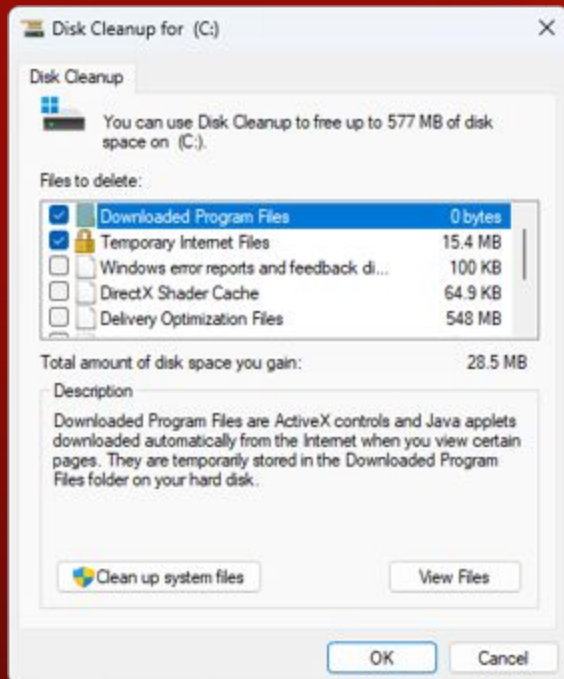
# Antivirus and Security Utilities

- ◆ Protect against malware and viruses
- ◆ Examples: Avast, Kaspersky, Windows Defender



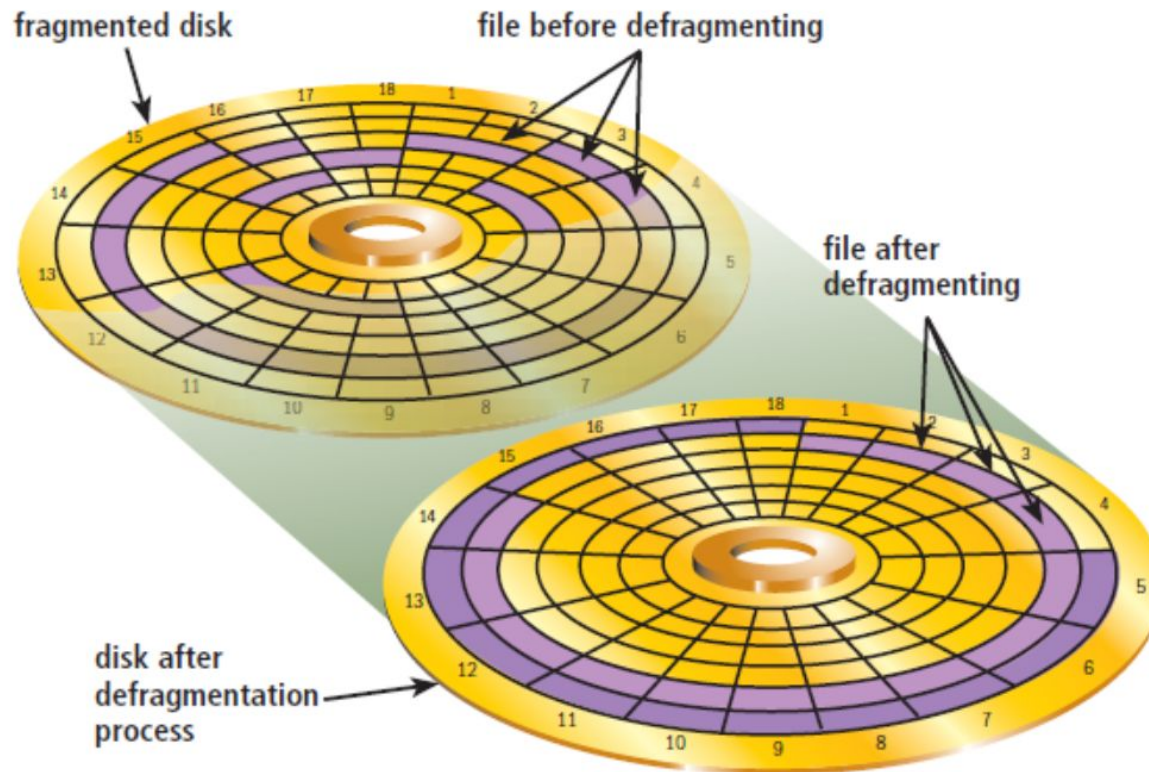
# Disk Management Utilities

- ◆ Disk Cleanup: Removes temporary/unwanted files
- ◆ Disk Defragmenter: Reorganizes fragmented data for faster access
- ◆ Partition Managers: Create/resize/delete disk partitions



# Disk Management Utilities

- ◆ Disk Defragmenter: Reorganizes fragmented data for faster access

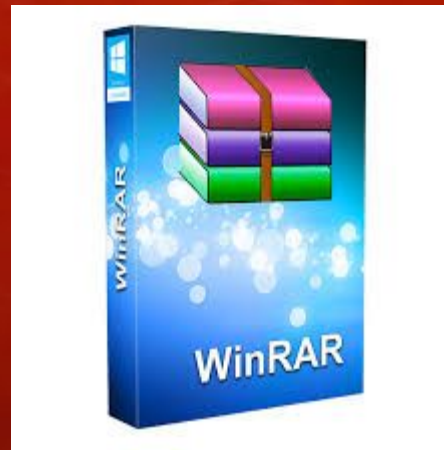
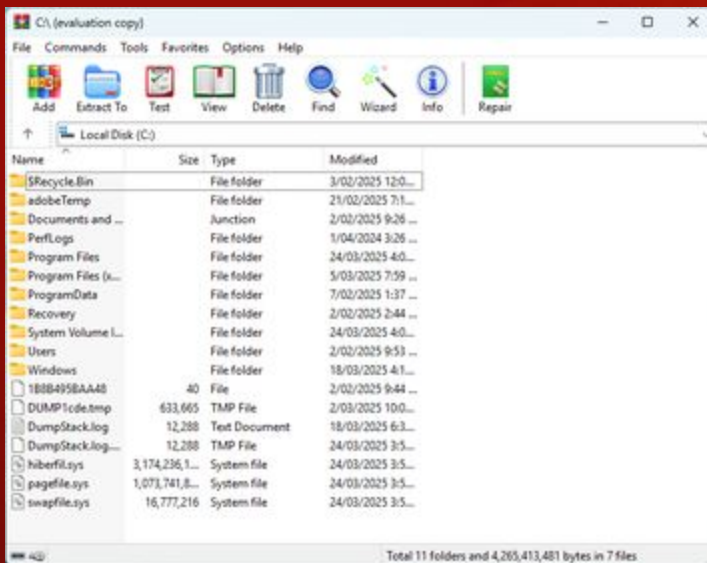


**Figure 7-18** A fragmented disk has many files stored in noncontiguous sectors. Defragmenting reorganizes the files so that they are located in contiguous sectors, which speeds access time.



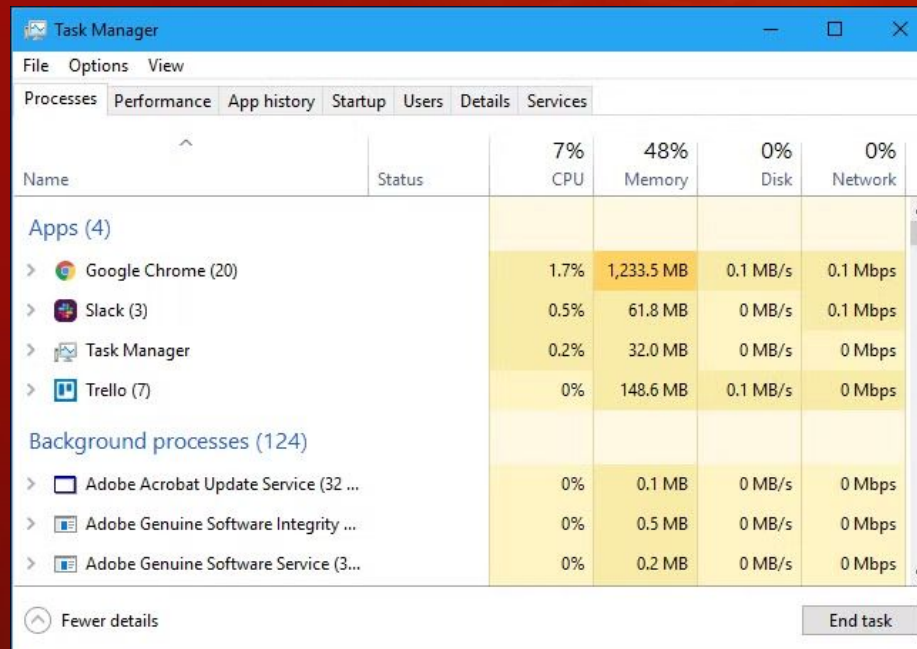
# Compression Utilities

- ◆ Reduce file size for storage & transfer
- ◆ Examples: WinRAR, WinZip, 7-Zip
- ◆ Used for archiving and sending files faster



# System Monitoring Utilities

- ◆ Task Manager, Resource Monitor
- ◆ CPU usage, memory usage, processes running
- ◆ Helps detect system performance issues



The screenshot shows the Windows Task Manager window with the 'Performance' tab selected. The window title is 'Task Manager'. The menu bar includes 'File', 'Options', and 'View'. The tabs at the top are 'Processes', 'Performance', 'App history', 'Startup', 'Users', 'Details', and 'Services'. The 'Performance' tab displays a summary of system resource usage: CPU at 7%, Memory at 48%, Disk at 0%, and Network at 0%. Below this summary is a table of running processes, categorized into 'Apps (4)' and 'Background processes (124)'. The table has columns for Name, Status, CPU, Memory, Disk, and Network. The 'Apps' section lists Google Chrome (20), Slack (3), Task Manager, and Trello (7). The 'Background processes' section lists Adobe Acrobat Update Service (32 ...), Adobe Genuine Software Integrity ..., and Adobe Genuine Software Service (3...). At the bottom of the window, there is a 'Fewer details' button and an 'End task' button.

| Name                                    | Status | 7% CPU | 48% Memory | 0% Disk  | 0% Network |
|-----------------------------------------|--------|--------|------------|----------|------------|
| <b>Apps (4)</b>                         |        |        |            |          |            |
| > Google Chrome (20)                    |        | 1.7%   | 1,233.5 MB | 0.1 MB/s | 0.1 Mbps   |
| > Slack (3)                             |        | 0.5%   | 61.8 MB    | 0 MB/s   | 0.1 Mbps   |
| > Task Manager                          |        | 0.2%   | 32.0 MB    | 0 MB/s   | 0 Mbps     |
| > Trello (7)                            |        | 0%     | 148.6 MB   | 0.1 MB/s | 0 Mbps     |
| <b>Background processes (124)</b>       |        |        |            |          |            |
| > Adobe Acrobat Update Service (32 ...) |        | 0%     | 0.1 MB     | 0 MB/s   | 0 Mbps     |
| > Adobe Genuine Software Integrity ...  |        | 0%     | 0.5 MB     | 0 MB/s   | 0 Mbps     |
| > Adobe Genuine Software Service (3...  |        | 0%     | 0.2 MB     | 0 MB/s   | 0 Mbps     |

# Other Utilities

- ◆ Firewall Utilities (prevent unauthorized access)
- ◆ Uninstall Managers (completely remove programs)
- ◆ Data Recovery Tools (restore lost files)
- ◆ Cloud Backup Utilities (Google Drive, OneDrive)



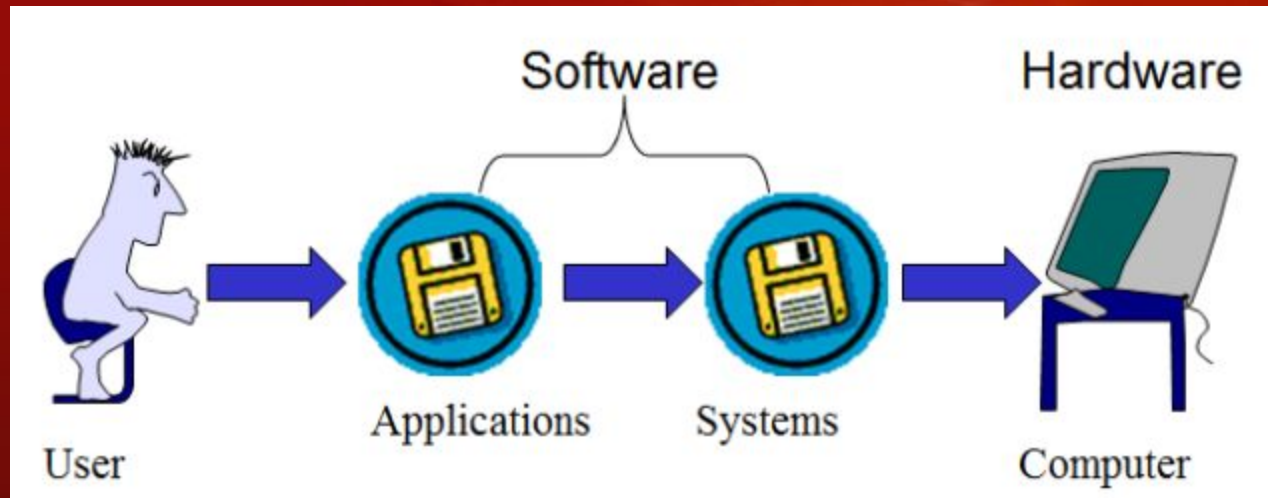
**FIREWALL**





# What is Application Software?

- ◆ Programs made for users (not for running the computer itself).
- ◆ Helps us do **specific** tasks:
  - Writing documents
  - Browsing internet
  - Watching videos
- ◆ Runs on **top of system software** (OS).



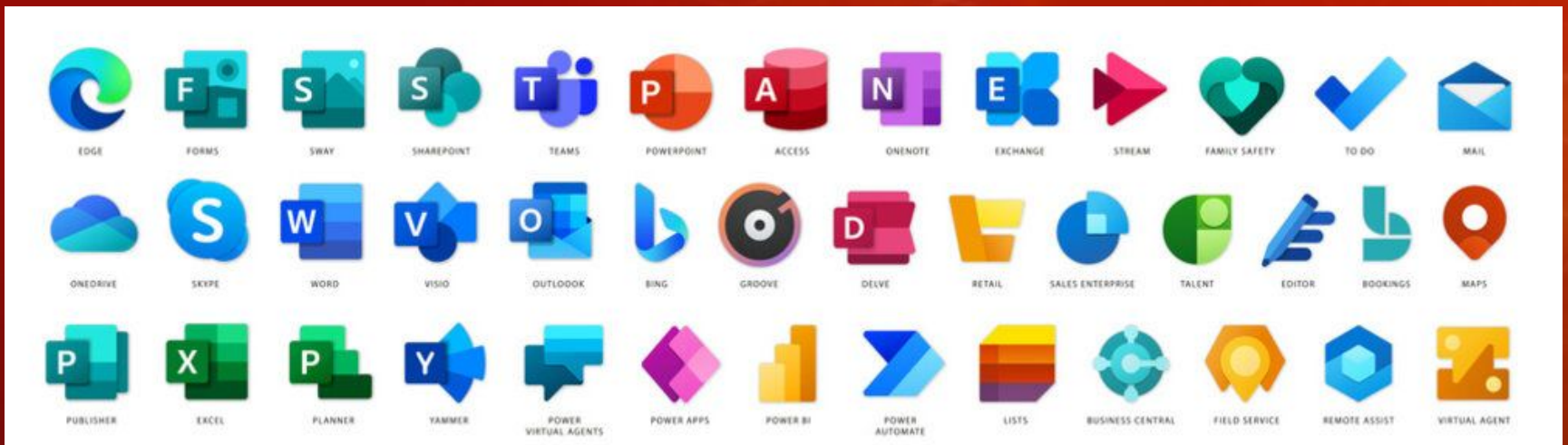
# What is Application Software?



**Figure 3-2** A user does not communicate directly with the computer hardware. Instead, system software is the interface between the user, the application software, and the hardware. For example, when a user instructs the application software to print a document, the application software sends the print instruction to the system software, which in turn sends the print instruction to the hardware.

# General Characteristics

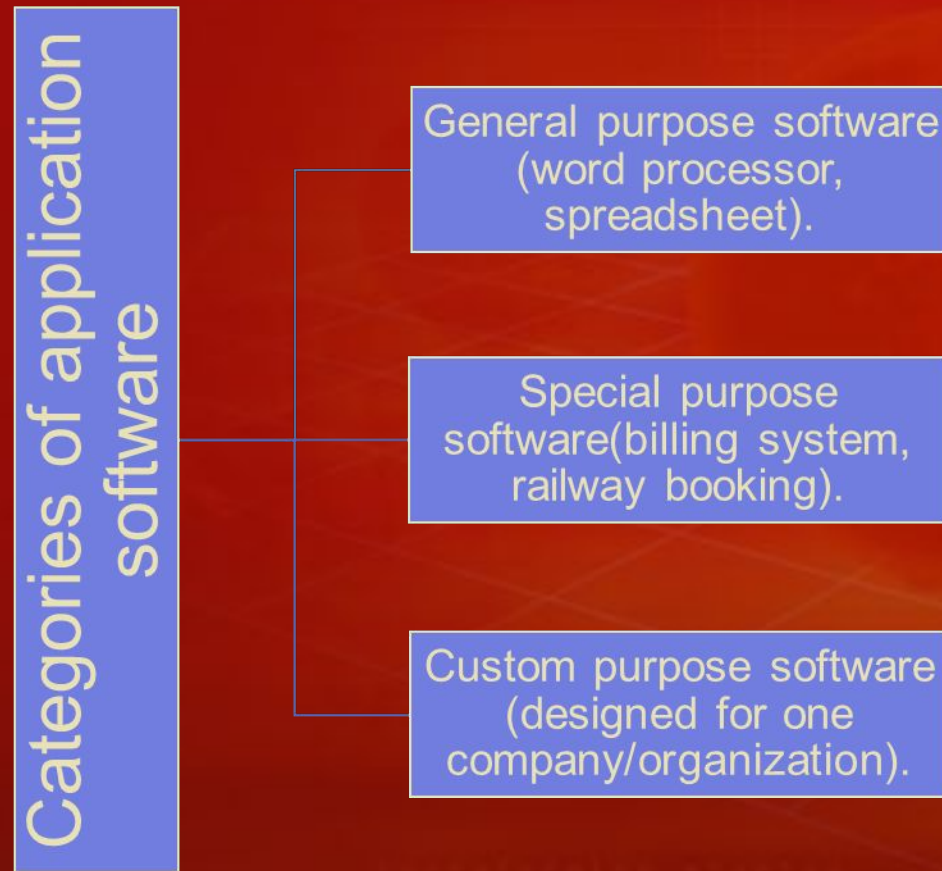
- ◆ User-oriented (solves user problems).
- ◆ Easy-to-use interface (GUI-based).
- ◆ Task-specific (e.g., word editing, calculations).
- ◆ Requires system software to run.
- ◆ Updated frequently with new features.



# Examples of Application Software

| Example         | APPs                    |
|-----------------|-------------------------|
| Office tools    | Word, Excel, PowerPoint |
| Multimedia apps | Photoshop, VLC Player   |
| Internet tools  | Browsers, Zoom, Teams   |
| Mobile apps     | WhatsApp, TikTok        |

# Categories of Application Software



# General-Purpose Software

- ◆ Used by everyone.
- ◆ Common tasks: typing, calculation, presentations.
- ◆ Examples:
  - Word processors
  - Spreadsheet software
  - Browsers

# Special-Purpose Software

- ◆ **Made for specific tasks.**
- ◆ **Not used by all users.**
- ◆ **Examples:**
  - **Hotel management system**
  - **Hospital patient system**
  - **Library software**



# Custom Software

- ◆ **Developed for a particular organization.**
- ◆ **Tailored to their requirements.**
- ◆ **Examples:**
  - **Airline reservation systems**
  - **University student portal**
  - **ERP software for a company**

# Packaged vs Custom Software

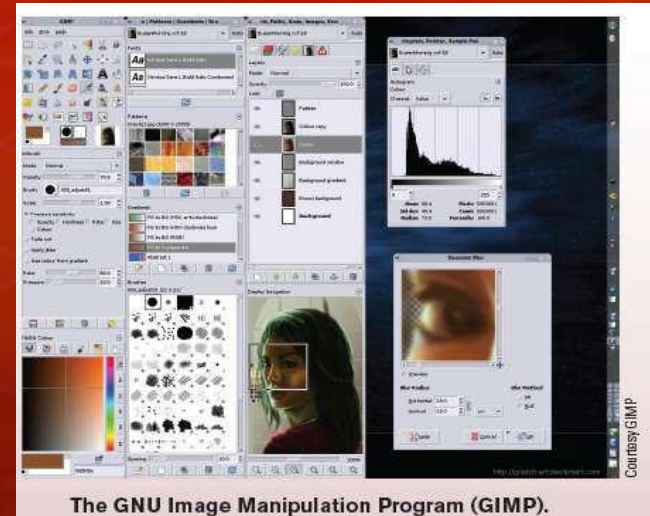
| Feature  | Packaged Software | Custom Software |
|----------|-------------------|-----------------|
| Made for | General public    | One client      |
| Cost     | Cheaper           | Expensive       |
| Example  | MS Office         | University ERP  |

# Software Ownership Rights

- ◆ Specify how a program can be used.
- ◆ Software License:
  - Legal right to use the program.
  - Defines conditions for buyer's use.

# Open Source Software

- ◆ Source code available to public.
- ◆ Growing use in industry.
- ◆ Typically cheaper, more stable, and secure.
- ◆ Examples:
  - Linux, LibreOffice, MySQL, GIMP.



# The Basics of Application Software

## ◆ **Commercial Software**

- Copyrighted software developed and sold for profit
- Typically comes with a single-user license

## ◆ **Shareware**

- Copyrighted software distributed on the honor system
- Consumers should either pay for it or uninstall it after the trial period

## ◆ **Freeware**

- Copyrighted software programs that are given away by the author for others to use free of charge

## ◆ **Public Domain Software**

- Software that is not copyrighted and ownership rights have been donated to the public domain

# The Basics of Application Software

## TYPE OF SOFTWARE

## EXAMPLES

Commercial software

Microsoft Office (office suite)  
Norton AntiVirus (antivirus program)  
Adobe Photoshop (image editing program)  
Minecraft - Pocket Edition (game)

Shareware

WinZip (file compression program)  
Video Edit Magic (video editing program)  
Image Shrinker (image optimizer)  
Deluxe Ski Jump 3 (game)

Freeware

Chrome (Web browser)  
LibreOffice (office suite)  
QuickTime Player (media player)  
Evernote (notetaking/archiving software)

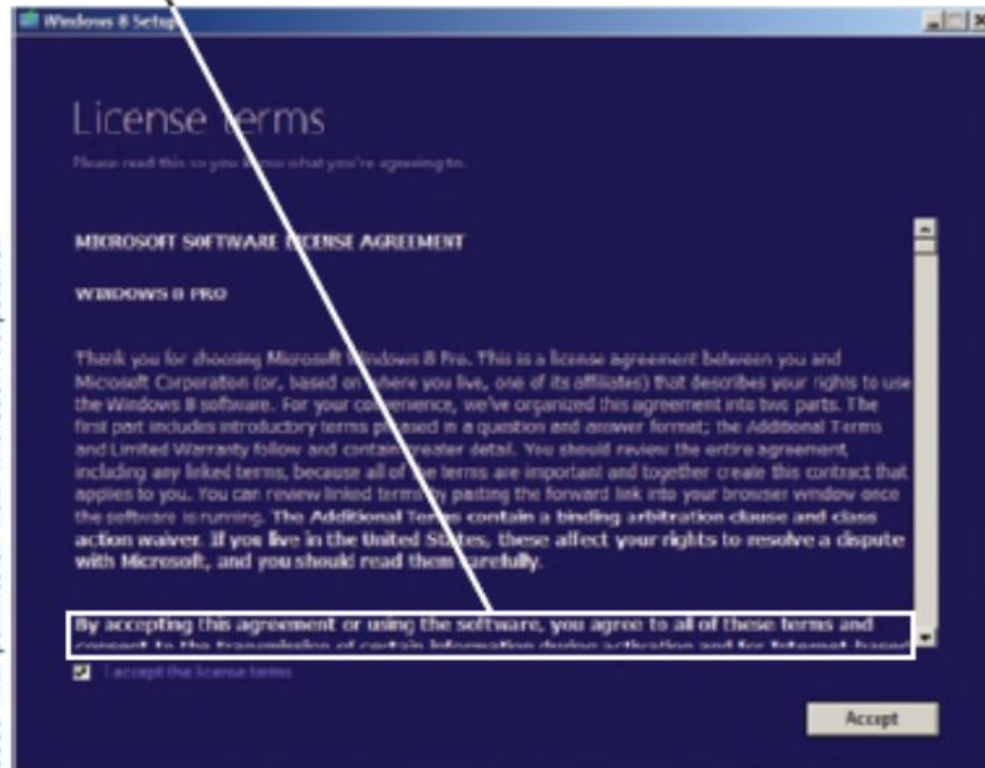
Public domain software

Lynx (text-based Web browser)  
Quake 3 (game)

# The Basics of Application Software

This statement explains that you are accepting the terms of the license agreement by installing the software.

Used with permission from Microsoft Corporation

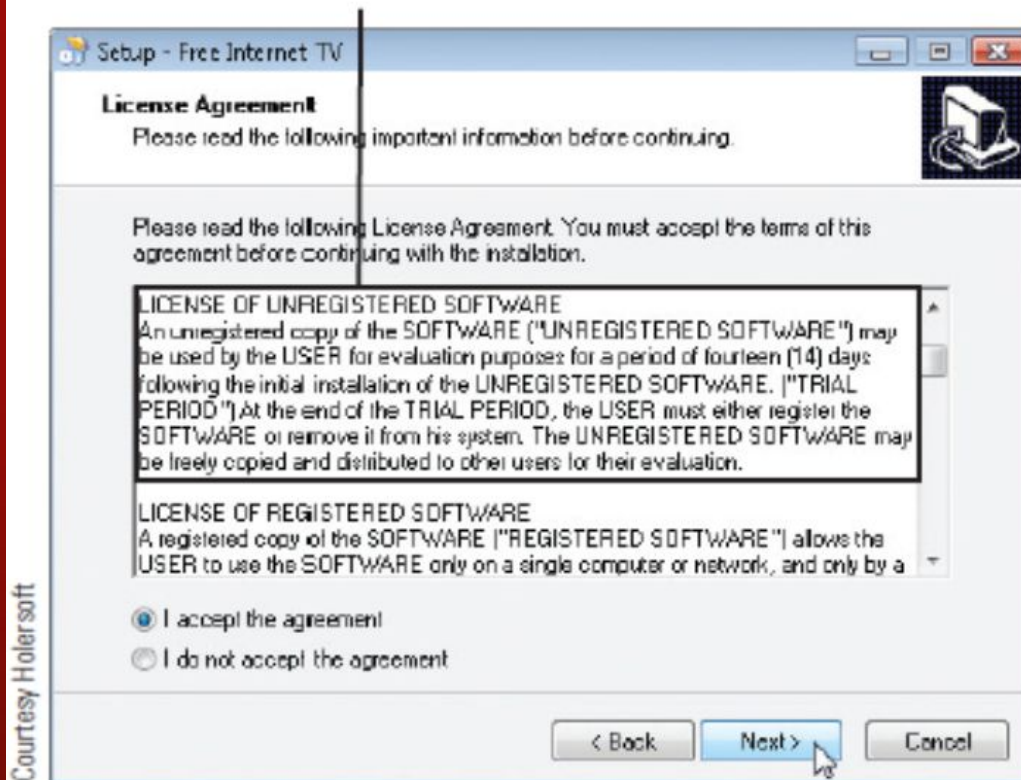


COMMERCIAL SOFTWARE PROGRAM



# The Basics of Application Software

This statement explains that the program can be tried for 14 days and then it needs to be either registered or uninstalled.



**SHAREWARE PROGRAM**

# Freeware vs Shareware vs Open Source

- ◆ **Freeware:**
  - Free forever (VLC Player).
- ◆ **Shareware:**
  - Free trial, then paid (WinRAR).
- ◆ **Open Source:**
  - Free + code available (LibreOffice, Linux apps).

# Productivity vs Entertainment Apps & Pros/Cons of Application Software

| Category           | Examples              | Purpose                               |
|--------------------|-----------------------|---------------------------------------|
| Productivity Apps  | Word, Excel, Zoom     | Help in work, study, communication    |
| Entertainment Apps | VLC, Spotify, Netflix | Provide fun, media, music, relaxation |

## ◆ Advantages ✓

- Makes tasks easier
- Saves time and effort
- Improves productivity
- Easy to learn and use

## ◆ Limitations ✗

- Need more system resources
  - (RAM, CPU)
- Can be expensive
- May require frequent updates
- Some apps depend on internet

# Summary

- ◆ Revise for Midterm

# Suggested Reading

- ◆ All my lectures and recommended book chapters